

Exploratory Performance Analysis on NLP Features from TV Series Scripts using Subgroup Discovery

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Figure 1: An Image of the Warner Bros. TV Series *Friends* [1]. It depicts the TV Series that the method is deployed on.

Abstract

This paper applies Subgroup Discovery to Natural Language Processing (NLP) features to uncover localized linguistic patterns in text. Engineered features such as humor ratios, sentiment shifts, and location references allows for the approach to find meaningful subgroups. The method is applied to the *Friends* sitcom script to explore combinations of NLP features that correlate with audience ratings. Results reveal that episodes set outside central locations with strong main character interactions, concise dialogue, and minimal minor character involvement receive higher ratings, while episodes confined to familiar settings with dense humor and fewer main interactions under-perform. These results highlight the value of Subgroup Discovery for interpretable, context-aware text analysis, with potential applications in entertainment, customer service, and resume evaluation.

Keywords

Subgroup Discovery; Natural Language Processing; Sentiment Analysis; Humor Detection; Emotion Detection; Performance Analysis

1 Introduction

In recent years, sitcom production has declined sharply as streaming platforms favor drama-oriented series with fewer yet longer episodes [2, 3]. However, audience data reveal that older and longer sitcoms such as *Friends* and *The Big Bang Theory* remain highly popular, accounting for 5.5 - 15.7% of U.S. streaming demand [4]. Whether for nostalgia or the variety of episodes to binge-watch, viewers return to these shows, suggesting the genre's enduring appeal. This presents a clear opportunity for production companies: creating new sitcoms could attract a loyal audience and generate profit. However, to maximize success and minimize production risk, directors must understand which narrative and stylistic features consistently resonate with audiences.

This paper addresses that challenge by analyzing a sitcom's script to identify the features that contribute to a well-received episode. Scripts contain many interacting choices, such as setting, tone, and character balance, resulting in a great number of possibilities for analysis. By developing interpretable descriptions that connect script characteristics to audience perception, production companies can make informed decisions about the features that lead to a highly-regarded episode before production begins.

Current practice relies primarily on expert judgment, script revisions, table reads, and limited audience testing [5]. If any data are used, they typically come from post-hoc analytics, as suggested in [6], offering little guidance at the writing stage on what features result in stronger episodes. As a result, decisions remain manual, time-consuming, and difficult to generalize across seasons or shows.

A promising approach is to summarize successful trends through descriptive patterns tied to recognizable script attributes. Such patterns can guide concrete writing and editing choices while helping stakeholders reason about aspects that consistently appear in higher-rated episodes or combinations that tend to under-perform.

Subgroup Discovery (SD) [7] is well-suited to this objective. SD identifies subpopulations of episodes that share simple, human-readable descriptions and differ significantly in audience ratings compared to other episodes. When applied to script-derived attributes, such as lexical, structural, or scene-composition indicators, SD can produce concise rules that reveal how particular script characteristics relate to stronger or weaker episodes.

As a case study, the script of *Friends* [8] is used. Several subgroups of episodes emerge, described using attributes derived from Natural Language Processing (NLP) feature engineering on the scripts. These descriptions are actionable: they map to features available to producers and writers, such as location distribution and dialogue sentiment, and are interpretable and specific enough to guide early-stage script development decisions.

This analysis contributes to the fields of SD, NLP, and entertainment analytics. The contribution includes: 1) the combination of NLP feature engineering and SD to create an interpretable descriptive framework linking script features to audience reception; 2) the application of this framework to television scripts at scale, incorporating within-episode dynamics, namely averages, shares, and peaks, to connect them to overall reception; and 3) the discovery of subgroups of episodes with distinct audience reception patterns.

2 Background and Related Work

A few studies [9–11] have focused on using Natural Language Processing (NLP) to analyze script or social media content to predict commercial success. Often, these cases center around movie scripts, specifically mainstream Hollywood movies [12, 13]. They apply techniques such as word frequency analysis, word embeddings, text preprocessing and tokenization, and sentiment or emotions analysis. For example, [13] attempted to combine various machine learning (ML) models with NLP features to predict box office revenue.

There are relatively few studies that target TV shows, specifically sitcoms. A 2017 study [14] focused on TV series pilot scripts of over 500 series in various genres to predict their IMDb ratings. The paper utilized various NLP techniques, including stopword removal, Part-of-Speech (POS) Tagging, Named Entity Recognition (NER), location frequency and proportion extraction using Conditional Random Fields (CRF), and dependency parsing between entities, such as characters and locations. These features were used as input to classical machine learning classifiers.

Despite the extensive extraction of NLP features from movie scripts and reviews [15], academic research often relies on a limited set of linguistic or emotional attributes. For example, interactions between characters or dialogue shifts are either forgotten about

or are only discussed at surface-level [16]. Moreover, sentiment scores are generally treated as uniform across text, seen by the usage static embeddings [17, 18]. This approach fails to capture complex emotional temporal regression, i.e. how emotions and/or topics evolve across time. One exception to the norm is a paper [19] which performed sentiment analysis on movie scripts. They extracted the sentiment of temporally subsequent parts of the movie, and analyzed it visually. However, this lack of advanced analysis methods beyond visual inspection poses a gap within research.

A limitation of existing ML approaches is their global optimization bias. ML models tend to prioritize predictive accuracy over interpretability, making it difficult to understand exactly why certain features are associated with success.

2.1 Local Pattern Mining in Text Analysis

Current literature has explored localized, context-aware methods known as Local Pattern Mining (LPM) [20]. These methods uncover local patterns linking specific textual configurations to ratings, rather than relying on global averages. This motivates combining LPM with NLP features. Key LPM techniques include exceptional model mining [20] and subgroup discovery [7].

Subgroup discovery is a data mining method that identifies distinctive relationships between a target variable and explanatory variables, focusing on revealing subpopulations with notable target-related patterns rather than global classifications.

One paper [21] took a similar approach using Named Entity Recognition (NER), Linking (NEL) and SD to analyze sentiment in Slovenian news articles. The authors first used Bidirectional Encoder Representations from Transformers (BERT) to identify named entities and linked to their corresponding entries in Wikipedia. However, this paper focuses solely on sentiment analysis.

To address the shortcomings of existing NLP techniques and the current NLP and SD combination mentioned above, the project starts by engineering a set of innovative features that add novelty to this paper in addition to the novel combination of SD with NLP.

3 Feature Engineering

To model and explain variation in external outcomes, a comprehensive set of linguistic, emotional, and structural *descriptors* is derived from line-structured narrative text. The corpus-agnostic approach applies to any document composed of ordered lines, such as screenplays, subtitles, podcast transcripts, or book chapters.

Using standard Natural Language Techniques techniques, raw text is transformed into quantitative descriptors capturing each unit’s narrative and style. These include lexical and stylistic measures based on word usage, sentiment and emotion indicators for affective tone, humor-related metrics for comedic tone, interaction measures describing conversational dynamics, and location distributions summarizing scene settings. Together, these descriptors serve as inputs to the Subgroup Discovery framework, enabling the identification of units that exhibit exceptional behavior relative to a target variable such as audience ratings.

Data and notation. Let the data set be $\Omega = \{r_1, \dots, r_N\}$, where each record r is a narrative unit (e.g., an episode) composed of $L(r)$ ordered lines $r = \{d_1, \dots, d_{L(r)}\}$. Each line d_i may include metadata

such as $\text{speaker}(d_i)$ and $\text{location}(d_i)$. The feature extractor

$$\phi : \Omega \rightarrow \mathbb{R}^k, \quad \mathbf{x}_r = \phi(r)$$

maps each unit to a k -dimensional vector of scalar descriptors (defined below). Each unit r is thus represented by $\mathbf{x}_r$ and paired with a target variable $y_r \in \mathbb{R}$ (e.g., a rating measure), forming a supervised data set $\{(\mathbf{x}_r, y_r)\}_{r \in \Omega}$.

3.1 Lexical and Stylistic Features

To capture linguistic and stylistic variation, surface-level features can be derived from text data. These quantify aspects such as dialogue length, lexical diversity, and speaker participation.

Definition 3.1 (Average words per line). For a record r with $L(r)$ lines and word counts $w_1, \dots, w_{L(r)}$,

$$\bar{w}(r) = \frac{1}{L(r)} \sum_{i=1}^{L(r)} w_i.$$

Higher values mark longer utterances; lower ones, faster pacing.

Definition 3.2 (Vocabulary richness). Let $T(r)$ be the total number of word tokens in record r and $U(r)$ the number of unique types.

$$\text{Vocabulary richness}(r) = \frac{U(r)}{T(r)}.$$

Higher values reflect greater vocabulary richness and variation.

Definition 3.3 (Main character participation share). Let C be the set of main characters. For each $c \in C$, let $L_c(r)$ be the number of lines spoken by c in record r , and $L(r)$ the total number of lines.

$$s_c(r) = \frac{L_c(r)}{L(r)}.$$

This gives the share of dialogue by each main character.

Definition 3.4 (Other characters' participation share). The share of all non-main characters is

$$s_{\text{OTHER}}(r) = 1 - \sum_{c \in C} s_c(r),$$

representing their collective dialogue contribution.

3.2 Sentiment and Emotion Features

Sentiment- and emotion-based features capture each record's affective tone and dynamics, quantifying overall polarity and the prevalence of specific emotions to reveal how affective patterns may relate to audience reception.

Definition 3.5 (Emotion score). These features capture the relative intensity of affective states, providing a fine-grained emotional profile for each record. Using a pretrained emotion classification model from Hugging Face [22], each line d_i receives scores for seven emotions, namely $e \in \{\text{joy, sadness, anger, fear, surprise, disgust, neutral}\}$. Let $s_{i,e}$ denote the score for emotion e on line i . The record-level proportion is then computed as

$$p_e(r) = \frac{1}{L(r)} \sum_{i=1}^{L(r)} s_{i,e}.$$

Definition 3.6 (Average sentiment polarity). Each record's sentiment is estimated with the VADER analyzer [23], which assigns four polarity scores to each line: negative (*neg*), neutral (*neu*), positive (*pos*), and a composite (*compound*). Let $z_{i,s}$ denote the score for $s \in \{\text{neg, neu, pos, compound}\}$ on line i ; the record-level mean is

$$m_s(r) = \frac{1}{L(r)} \sum_{i=1}^{L(r)} z_{i,s}.$$

neg, *neu*, and *pos* range from 0–1, describing proportions of negative, neutral, and positive affect. *compound* ranges from −1 (negative) to +1 (positive), summarizing overall affective balance.

Definition 3.7 (Sentiment trajectory slope). To capture emotional progression within a record, the VADER compound score [23] is computed per line and smoothed with a moving average. Let $x_i \in [0, 1]$ be the normalized line position and y_i the smoothed sentiment. A linear model

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$$

is fitted, where the slope $\beta_1(r)$ quantifies sentiment drift: $\beta_1 > 0$ indicates a positive trend, $\beta_1 < 0$ a negative one.

3.3 Humor Related Features

Humor is a common most in most movies or series. Quantifying it allows analysis of its relation to audience engagement and its role in narrative style variation.

Definition 3.8 (Humor share). Each dialogue line is evaluated using a pretrained humor detection model [24], which assigns a binary label for humor. Let $H(r)$ be the number of humorous lines in record r and $L(r)$ the total number of lines. The proportion of humorous dialogue is then defined as

$$h(r) = \frac{H(r)}{L(r)}.$$

This feature captures the relative density of humor, reflecting how much the dialogue relies on comedic versus narrative content.

3.4 Character Interaction Features

Character interaction features examine how characters engage throughout the text. They quantify how frequently specific character duos alternate in conversation, revealing relationship dynamics and interaction balance. Such patterns influence audience perception, as recurring exchanges between familiar pairs can enhance emotional engagement and narrative coherence.

Definition 3.9 (Dialogue exchanges by pair). For a character pair $(c_p, c_q) \in C \times C$, an *exchange* is two directly alternating lines where c_p and c_q speak consecutively. Let $s_i = \text{speaker}(d_i)$ denote the speaker of line i . The total exchanges in a record r are

$$E_{c_p, c_q}(r) = \sum_{i=2}^{L(r)} \mathbf{1}\{(s_{i-1}, s_i) \in \{(c_p, c_q), (c_q, c_p)\}\}.$$

This feature quantifies how often specific character relationships appear in a text. Records with a high number of exchanges for a particular pair indicate a narrative focus on that relationship, whereas lower counts reflect more distributed conversational structures.

3.5 Location Features

Location features capture where dialogue unfolds, linking scene setting to the text’s overall composition. They measure the proportion of dialogue in each location, reflecting shifts between settings. This spatial distribution shapes narrative rhythm, visual diversity, and the audience’s sense of setting and continuity.

Definition 3.10 (Location share). To construct location-based features, scene metadata is extracted from scene notes using regular expressions. Frequent variations in location naming are normalized into canonical forms and grouped into a fixed set of categories L , depending on the data set. Let $\text{location}(d_i) \in L$ denote the location associated with dialogue line i . For each location $l \in L$, the share of dialogue occurring in that location within a record r is defined as

$$\pi_l(r) = \frac{1}{L(r)} \sum_{i=1}^{L(r)} \mathbf{1}\{\text{location}(d_i) = l\}.$$

Each $\pi_l(r)$ constitutes an individual feature representing the proportion of dialogue taking place in location l , collectively describing how the record’s content is distributed across different settings.

4 Subgroup Discovery Framework

In traditional Subgroup Discovery (SD), the goal is to find subsets of the data that show statistically significant deviations in the target variable compared to the overall population or the complement of the subset [7]. These subgroups are defined by simple descriptive conditions over the attributes of the data set, such as combinations of categorical and numerical constraints, that make the subgroup both interpretable and distinctive.

To efficiently explore the search space of all possible combinations of subgroups, beam search is used. It is a heuristic search method that can balance greedy search, which may be fast yet has no guarantee of finding the optimal solution, and computationally expensive exhaustive search. Beam search is extensively used in the field of NLP to perform tasks such as machine translation, text summarization and speech recognition. For further clarification, refer to the algorithm’s pseudocode as proposed in [20].

4.1 Quality Measures

There can be millions of supposedly interesting subgroups without proper scoring functions. They are needed to quantify how “interesting” or useful the patterns are and what interesting means for the specific application. The quality of candidate subgroup can be assessed by how different the subgroup’s target distribution is from the overall population, and the size of the subgroup. A number of potentially interesting quality measures are used for this analysis.

Definition 4.1 (Welch’s t-test).

$$t = \frac{\mu_{\text{sub}} - \mu_{\text{comp}}}{\sqrt{\frac{s_{\text{sub}}^2}{n_{\text{sub}}} + \frac{s_{\text{comp}}^2}{n_{\text{comp}}}}}$$

s_{sub}^2 : variance of subgroup

s_{comp}^2 : variance of complement

μ_{comp} : mean of complement

Computes the significance of the mean difference between a subgroup and its complement or the population, accounting for variance and sample size [25].

Definition 4.2 (Kullback–Leibler Coverage-Sensitive Weighting (KLCSW)). KLCSW weights a subgroup’s mean difference by how close its coverage c is to a target c_0 :

$$D_{\text{KL}}(c \parallel c_0) = c \ln \frac{c}{c_0} + (1 - c) \ln \frac{1 - c}{1 - c_0},$$

$$w_{\text{KLCSW}}(c) = e^{-\gamma D_{\text{KL}}(c \parallel c_0)},$$

$$S_{\text{KLCSW}} = w_{\text{KLCSW}}(c) \cdot |\mu_{\text{sub}} - \mu_{\text{comp}}|.$$

Here, c_0 is the target coverage and γ controls weighting sharpness. KLCSW ensures subgroups are neither too small nor too large relative to c_0 [26].

Definition 4.3 (Beta Coverage Weighting (BKW)). BKW extends standard feature weighting methods by introducing a coverage weight shaped by a Beta distribution with parameters α and β [27]. This skews the entropy function, ensuring subgroups are neither too small nor too large while giving similarly interesting-sized subgroups comparable weight.

$$w_{\text{BKW}}(c) = \frac{c^\alpha (1 - c)^\beta}{\left(\frac{\alpha}{\alpha + \beta}\right)^\alpha \left(\frac{\beta}{\alpha + \beta}\right)^\beta},$$

$$S_{\text{BKW}} = w_{\text{BKW}}(c) \cdot |\mu_{\text{comp}} - \mu_{\text{sub}}|.$$

For all quality measures, the subgroup can be compared either to its complement or to the population. The choice depends entirely on the use case and whether the chosen or interesting subgroups are expected to have a relatively high coverage or not.

5 Experiments

To deploy the Subgroup Discovery (SD) framework on the feature-engineered Natural Language Processing (NLP) features, the analysis uses data from the *Friends* sitcom [8]. The sitcom was chosen for its popularity, high ratings, and presence on binge-watched lists [2, 3]. The show’s lengthy run of ten seasons provides a sufficient amount of data. *IMDb* scores were chosen to measure audience ratings. The code for the project is available online on GitHub ¹.

5.1 Data Description

For this research paper, two data sets regarding the *Friends* sitcom were used. Both data sets are available online at [28] and [29].

The script data set from [28] consists of 70369 rows and 14 columns, representing the complete *Friends* script line by line. Each row includes the season, episode, and title, with the credited writers. Lines are categorized into three types: scene note, stage direction and dialogue. Scene notes describe the setting or context; stage direction detail environmental or character actions; and dialogue lines contain spoken text, with and without embedded stage directions. Each dialogue line is associated with a speaker, which may be a single person or a group of people speaking in unison. Together, these lines form the full textual structure of each episode’s script.

¹GitHub repository:

<https://github.com/DimitarDimitrov2001/EMM-Friends-Project>

The ratings data set from [29] includes 236 rows and 9 columns, corresponding to all *Friends* episodes. Each row provides the episode title, season, year, duration (usually 22-30 minutes), a one-sentence summary, director, IMDb vote count, and IMDb star rating, ranging from 1 - 10, which serves as the target variable for analysis.

5.2 Data Preprocessing

The preprocessing phase aims to combine the two data sets. Episode identifiers were standardized and two-part episodes were manually split to ensure alignment between the data sets. Non-dialogue rows such as voice-over recaps and headers were removed leaving only meaningful text; dialogue, scene notes, and stage directions. Scene notes were further processed to extract and normalize primary locations, which were aggregated into broader categories to calculate location share per episode (see subsection 3.5).

The script data set was then restructured at the episode level for feature engineering. Line-level data were preserved to compute character-level dialogue metrics and interaction patterns, which were later aggregated by episode. Finally, the processed scripts were merged with IMDb episode ratings using the standardized season and episode keys. Feature engineering on the script derived columns described in section 3; for character-level NLP features, the six protagonists were used. The final data set contains 236 rows for each episode and 26 columns with the additional NLP features, forming the basis for subsequent analysis.

5.3 Experimental Setup

The analysis applies Subgroup Discovery via single-beam search with width 2000 and depth 3, exploring conjunctions of simple, interpretable conditions over the derived attributes. Numerical attributes are split into 5 parts for the subgroups of the first layer. These parts are subsequently used in the creation of higher order subgroups without further refinement. Quality is assessed using three measures for mean differences on a continuous target: Welch’s t-statistic [25], Beta coverage-weighted mean difference (BKW) [27] with $\alpha = 0.43$ and $\beta = 1$, and Kullback–Leibler coverage-sensitive weighting (KLCSW) [26] with $c_0 = 12$ and $\gamma = 12$.

Subgroups are compared with their complements to avoid comparing largely to themselves. BKW and KLCSW parameters were tuned to reduce bias toward very small or large subgroups: small subgroups have high variance, and large ones dominate if coverage weighting is too strong. Parameters were selected to yield generic, diverse, and meaningful subgroups. Each measure is run twice, for exceptionally high (maximum) and low (minimum) mean ratings.

Moreover, a manual review of the discovered subgroup descriptions was performed before any interpretation. Descriptions that were judged “too generic” (for example those composed primarily of common conditions, default ranges, or predicates without meaningful restrictions) were removed. This way, false discoveries were removed. The remaining subgroup descriptions provide specific, discriminative characterizations of episodes rather than broad summaries, and only these were used as results.

From the 25 subgroups generated for each run (each quality metric either maximum or minimum), the highest rated ones were interpreted first and with the most detail, while the rest were analyzed more superficially.

5.4 Results

In Table 1, the top two subgroups for each quality measure (one for high-mean, one for low-mean) can be seen, along with summary information on the subgroups. Moreover, the rest of the subgroups can be seen in the Appendix in Table 2.

Table 1: Global overview of SD runs: top subgroup of 25 subgroups per run, ranked by quality measure score.

Run	Best Quality Description	Size	μ_{sub}	μ_{comp}	Q
Welch (max)	<i>Central Perk</i> $\leq 0.05 \wedge$ <i>exchanges CHANDLER-JOEY</i> $\leq 31.67 \wedge$ <i>exchanges CHANDLER-PHOEBE</i> ≤ 5.0	16	8.93	8.42	8.30
Welch (min)	<i>exchanges CHANDLER-JOEY</i> $\leq 31.67 \wedge$ <i>exchanges MONICA-CHANDLER</i> $\leq 4.0 \wedge$ <i>exchanges ROSS-CHANDLER</i> > 7.0	18	8.1	8.49	10.73
BKW (max)	<i>Central Perk</i> $\leq 0.12 \wedge$ <i>avg_words_per_line</i> $\leq 9.1183 \wedge$ <i>character_lines OTHER</i> ≤ 0.16	13	9.08	8.42	0.43
BKW (min)	<i>character_lines PHOEBE</i> $\leq 0.1 \wedge$ <i>humor</i> $> 0.63 \wedge$ <i>vocab_richness</i> > 0.41	11	7.93	8.48	0.34
KLCSW (max)	<i>Central Perk</i> $\leq 0.17 \wedge$ <i>disgust</i> $\leq 0.05 \wedge$ <i>exchanges MONICA-CHANDLER</i> > 4	68	8.69	8.36	0.33
KLCSW (min)	<i>Central Perk</i> $> 0.17 \wedge$ <i>anger</i> $\leq 0.08 \wedge$ <i>exchanges CHANDLER-JOEY</i> ≤ 31.67	66	8.24	8.54	0.30

Note: Q stands for the quality score.

Across the three runs performed by the three quality measures (Welch’s t-stat, BKW, KLCSW) looking for *exceptionally high* or *maximum* mean ratings, some attributes were consistently present in subgroup descriptions:

- Lower presence of Central Perk was part of almost every subgroup description. Most descriptions bounded the presence to 5% or 12% of the entire episode. For example, Welch’s t-stat generated descriptions with *Central Perk* ≤ 0.0537 combined with constraints on character interactions and short lines or positive tone yielded subgroups with around 8.9 - 9.0 mean rating and a decent coverage (≈ 20 episodes). Same pattern also appears when BKW and KLCSW quality measure are used.
- Character interactions play a big part in the rating of an episode. Higher means often coincide with more exchanges between Chandler and Monica (e.g. *exchanges_MONICA_CHANDLER* > 4). This condition appears consistently between all three quality measures (most often using KLCSW), often accompanied by a low Central Perk share, but also lower interaction between other pairs (e.g. *exchanges_RACHEL_CHANDLER* ≤ 4).
- Higher mean ratings are often also results of shorter average lines (e.g. *avg_words_per_line* ≤ 10.5) and low disgust/positive tone (e.g. *emotion_joy* > 0.1135). These conditions appear mostly in subgroup descriptions generated by BKW and KLCSW, but also occasionally with t-stat.
- Subgroup descriptions with higher mean generated by the BKW and KLCSW measures often included higher presence Ross. Under BKW, this was mostly in combination with Rachel (e.g. *exchanges_RACHEL_ROSS* > 12), while under KLCSW, a lot of descriptions simply included a higher number of lines said by Ross (e.g. *character_lines_ROSS* > 0.1034).

Overall, the most common high-mean description is that which includes: small share of Central Perk *and* high Monica-Chandler exchanges *and* simple, concise lines.

Similarly, across the three runs performed by the quality measures looking for exceptionally small mean ratings, a few attributes emerge as quite often in the subgroup descriptions. Unsurprisingly, these attributes were often the same as the ones present in high-mean descriptions, but mirrored:

- Higher presence of Central Perk within an episode usually accompanies a lower episode rating. The presence of Central Perk is mostly bounded at $> 16.67\%$. Combined with other factors like the number of certain character exchanges or lower joy, all three quality measures produce mean ratings of subgroups at around 8.1 - 8.2, which is well below the mean rating of the entire show, namely 8.48.
- Less Monica-Chandler exchanges and more exchanges including Ross often results in a worse rated episode. T-stat produces several subgroup descriptions with $exchanges_MONICA_CHANDLER \leq 4$ and $exchanges_ROSS_CHANDLER > 7$ or > 11 which have exceptionally low mean rating while having a decent coverage (≈ 20). Similarly, descriptions generated by KLCSW often include $exchanges_ROSS_JOEY > 2$ paired with higher share of Central Perk.
- KLCSW measure also often generates lower-mean subgroup descriptions which include lower amount of angry lines (eg. $emotion_anger \leq 0.079$) or minimal presence of hallway scenes (eg. $Hallways \leq 0$), usually paired with higher presence of Central Perk. On the other hand, t-stat and BKW measures produce a lot of lower-mean subgroups with longer lines (eg. $avg_words_per_line > 10.53$) and richer vocabulary (eg. $vocab_richness > 0.3865$), but again usually paired with high Central Perk presence.
- Presence of the main apartments plays a big role in lower-mean subgroups discovered by all three quality measures. However, BKW significantly differs from t-stat and KLCSW. While subgroups descriptions discovered by BKW often include a higher presence of the main apartments (e.g. $Main_apartments > 0.4667$), lower-mean subgroups discovered by both t-stat and KLCSW actually include lower presence of the main apartments (e.g. $Main_apartments \leq 0.4667$).

Overall, the most common low-mean description is that which includes: high share of Central Perk *and* low amount of Monica-Chandler exchanges *and* often) longer lines with rich vocabulary.

The two most prevalent attributes among all six runs of the three measures were the percentage of the episode taking place in Central Perk (the lower, the better) and the amount of exchanges Monica and Chandler have within an episode (the more, the better). This suggests that the viewers enjoy seeing Monica and Chandler interact, which results in episodes such as Monica and Chandler's wedding or the one where Chandler, Joey, Monica and Rachel compete for the better apartment having higher ratings. Moreover, the viewers clearly enjoy the uniqueness of the main cast not spending time in Central Perk, since of the 40 episodes which have less than $\approx 5.3\%$ share of Central Perk, the vast majority has an above average rating. Interestingly, episodes with shorter average length of

lines tend to be higher rated while episodes with longer lines and more complex dialogue tend to be rated lower. This suggests that the viewers simply want to enjoy this type of show without having to focus on it too much and treat it more as an escape from reality than something they want to put their energy into.

6 Conclusion

Although sitcom production is declining, the genre remains popular for binge-watching [2, 3]. To help producers capitalize on this, we combine NLP features from episode scripts with Subgroup Discovery (SD) to identify subpopulations of episodes with high audience ratings, enabling early estimates of potential success.

The novelty lies in applying SD to NLP-derived proportional descriptors, such as humor or setting ratios and sentiment shifts, rather than standard frequency counts, and using the sentiment slope coefficient [19] as a feature over only visual inspection.

The strongest results from the *Friends* script analysis indicates that episodes set outside Central Perk, a central location within the series, with strong interactions among core characters, especially Chandler-Joey or Monica-Chandler, tend to receive higher ratings. Episodes with concise dialogue and minimal involvement of minor characters also perform well. Those that remain in Central Perk, feature fewer interactions between main characters, or are overly dense in humor and linguistic complexity tend to under-perform. Overall, audience enjoyment is maximized when episodes balance main character dynamics with engaging, snappy dialogue across newer and varied locations, guiding producers on which character relationships and interactions to emphasize.

Limitations include restricting character interactions to pairs even in group settings, analyzing only protagonists due to computational cost, and a narrow rating range (7.2-9.7), which reduces actionable differences. Applying the method to shows with broader rating distributions may yield stronger insights.

Multiple NLP models generated features such as prevalent emotions, sentiment scores, and humor ratios [22–24], are were evaluated on 12 manually annotated episodes (5% of the series) with 50-54% accuracy. The emotion model distinguished positive and negative emotions but confused specific classes; the sentiment model under-predicted neutral sentiment but had high recall for positive and negative; the humor model produced false negatives for humor absence and false positives for humor. These biases affect downstream subgroup interpretations.

6.1 Future work

Future research could expand NLP descriptors using Named Entity Recognition, dependency parsing, sarcasm detection [14], richer emotion-annotated dialogue, or manually annotated sarcasm datasets [30, 31]. Topic modeling, excluded here due to noisy data, may yield more informative subgroups. The single-beam SD algorithm could also be extended to multi-beam search [32] or evolve into Exceptional Model Mining (EMM) by incorporating multiple target variables, such as combining IMDb and Rotten Tomatoes ratings.

This work demonstrates integrating NLP with SD, demonstrated on sitcom scripts and is applicable to domains such as book recommendations, customer service transcripts, or resume analysis.

References

- [1] Warner Bros. Television. Friends, 2025. Retrieved October 1, 2025 from <https://www.warnerbros.canada.com/tv/friends>.
- [2] Daniel Parris. The Rise and Fall of TV Sitcoms: A Statistical Analysis, 2025.
- [3] Nielsen Media Research. Streaming unwrapped: Streaming viewership goes to the library in 2023, 2024.
- [4] Dade Hayes. "The Big Bang Theory" Is Nielsen's "Most-Binged" Streaming Title Of 2024, 2025.
- [5] Ken Aguado. How a TV Series Gets Made, 2022.
- [6] Andrea Fronzetti Colladon and Maurizio Naldi. Predicting the performance of TV series through textual and network analysis: The case of Big Bang Theory. *PLOS ONE*, 14(11):e0225306, 2019.
- [7] Francisco Herrera, Cristóbal José Carmona, Pedro González, and María José del Jesus. An Overview on Subgroup Discovery: Foundations and Applications. *Knowledge and Information Systems*, 29(3):495–525, 2010.
- [8] IMDb. Friends, 2025. Retrieved October 13, 2025 from <https://www.imdb.com/title/tt0108778/>.
- [9] Paschalis Frangidis, Konstantinos Georgiou, and Stefanos Papadopoulos. "Sentiment Analysis on Movie Scripts and Reviews". In Ilias Maglogiannis, Lazaros Iliadis, and Elias Pimenidis, editors, *Artificial Intelligence Applications and Innovations*, pages 430–438, Cham, 2020. Springer International Publishing.
- [10] Naisargi D. Sureja and Firoz A. Sherasiya. Using Sentimental Analysis Approach Review on classification of Movie Script, Apr 2017.
- [11] Nashit Ali1 Dewan Muhammad Qaseem1. Movie Success-rate prediction system through optimal sentiment analysis, 2022.
- [12] Suyanee Polsri, Ya-Wen Chang Chien, and Li-Chen Cheng. A machine learning model for predicting a movie sequel's revenue based on the sentiment analysis of consumers' reviews, Jan 1970.
- [13] Joshua A Gross, William C Roberson, and J Bay Foley-Cox. Cs 230: Film success prediction using nlp techniques. *Stanford University: CS 230 Course*, 2021.
- [14] Junshu Wang. *Information Extraction from TV Series Scripts for Uptake Prediction*. PhD thesis, Ph. D. thesis, Auckland University of Technology, 2017.
- [15] Paschalis Frangidis, Konstantinos Georgiou, and Stefanos Papadopoulos. Sentiment analysis on movie scripts and reviews: Utilizing sentiment scores in rating prediction. In *IFIP International Conference on Artificial Intelligence Applications and Innovations*, pages 430–438. Springer, 2020.
- [16] Myungji Lee, Hongseok Kwon, Jaehun Shin, Wonkee Lee, Baikjin Jung, and Jong-Hyeok Lee. Transformer-based screenplay summarization using augmented learning representation with dialogue information, 2021.
- [17] Wei Zhou and Jelke Bloem. Comparing contextual and static word embeddings with small data, Jan 1970.
- [18] Sérgio Barreto, Ricardo Moura, Jonnathan Carvalho, Aline Paes, and Alexandre Plastino. Sentiment analysis in Tweets: An assessment study from classical to modern word representation models, 2023.
- [19] Rifat Rahman, Md Abdul Masud, Raonak Jahan Mimi, and Mst Nusrat Sultana Dina. Sentiment analysis on adventure movie scripts. In *2020 2nd International Conference on Sustainable Technologies for Industry 4.0 (STI)*, pages 1–6. IEEE, 2020.
- [20] Wouter Duivesteijn, Ad J. Feelders, and Arno Knobbe. Exceptional model mining. *Data Mining and Knowledge Discovery*, 30(1):47–98, Feb 2015.
- [21] Anita Valmarska, Luis Adrián Cabrera-Diego, Elvys Linhares Pontes, and Senja Pollak. "Exploratory Analysis of News Sentiment Using Subgroup Discovery". In *Proceedings of the 8th Workshop on Balto-Slavic Natural Language Processing*, pages 66–72, Kiyv, Ukraine, April 2021. Association for Computational Linguistics.
- [22] Jochen Hartmann. Emotion English DistilRoBERTa-base, 2022. Retrieved October 10, 2025 from https://huggingface.co/michellejieli/emotion_text_classifier.
- [23] C. Hutto and Eric Gilbert. VADER: A Parsimonious Rule-Based Model for Sentiment Analysis of Social Media Text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1):216–225, 2014.
- [24] Mohamed Amine Dhiab. humor-no-humor, 2022. Retrieved October 10, 2025 from <https://huggingface.co/mohameddhiab/humor-no-humor>.
- [25] David Curtis. Welch's T test is more sensitive to real world violations of distributional assumptions than student's t test but logistic regression is more robust than either - statistical papers, Mar 2024.
- [26] Matthijs van Leeuwen and Arno Knobbe. Diverse subgroup set discovery - data mining and knowledge discovery, Jun 2012.
- [27] Hugo M Proença, Peter Grünwald, Thomas Bäck, and Matthijs van Leeuwen. Robust subgroup discovery: Discovering subgroup lists using mdl. *Data Mining and Knowledge Discovery*, 36(5):1885–1970, 2022.
- [28] Bree Nguyen and Blesson Densil. Friends TV/Television All Episode Dialogue CSV, 2025. Retrieved October 2, 2025 from <https://www.kaggle.com/datasets/brzy56/friends-tv-television-scripts-all-dialogue-csv>.
- [29] Mohammad Reza Ghari and Moulik Dhade. Friends Series Dataset, 2025. Retrieved October 2, 2025 from <https://www.kaggle.com/datasets/rezaghari/friends-series-dataset/data>.
- [30] Soujanya Poria, Devamanyu Hazarika, Navonil Majumder, Gautam Naik, Erik Cambria, and Rada Mihalcea. MELD: A Multimodal Multi-Party Dataset for Emotion Recognition in Conversations, 2019.
- [31] Aditya Joshi, Vaibhav Tripathi, Pushpak Bhattacharyya, and Mark Carman. *Harnessing Sequence Labeling for Sarcasm Detection in Dialogue from TV Series "Friends"*. Association for Computational Linguistics, 2016.
- [32] Ashwin K Vijayakumar, Michael Cogswell, Ramprasad R Selvaraju, Qing Sun, Stefan Lee, David Crandall, and Dhruv Batra. Diverse beam search: Decoding diverse solutions from neural sequence models. *arXiv preprint arXiv:1610.02424*, 2016.
- [33] IMDb. Friends: The stuff you've never seen, 2025. Retrieved October 13, 2025 from <https://www.imdb.com/title/tt0292518/>.
- [34] Parrot Analytics. The state of streaming and television in the United States Q3 2022: Netflix, NBC, CBS, HBO and ABC, 2022.
- [35] Rachael L Disantostefano and Keith E Muller. A comparison of power approximations for Satterthwaite's test, 1995.
- [36] Yuan Xu. Evaluation & Analysis of movie aspects: Based on sentiment analysis, 2023.

Table 2: Top five subgroup descriptions for the two runs (high- and low-mean) for each quality measure, ranked by the respective quality measure score.

Rank	Run	Description	Size	μ_{sub}	μ_{comp}	Q
1	T-stat (max)	IF Central Perk ≤ 0.0537 AND exchanges_CHANDLER_JOEY ≤ 31.6667 AND exchanges_CHANDLER_PHOEBE ≤ 5	16	8.93	8.42	8.30
2	T-stat (max)	IF Central Perk ≤ 0.0537 AND exchanges_RACHEL_CHANDLER ≤ 4	19	8.91	8.42	8.12
3	T-stat (max)	IF Central Perk ≤ 0.0537 AND Main apartments ≤ 0.6667 AND exchanges_RACHEL_CHANDLER ≤ 4	18	8.88	8.42	8.08
4	T-stat (max)	IF Central Perk ≤ 0.0537 AND exchanges_CHANDLER_JOEY ≤ 12 AND exchanges_RACHEL_CHANDLER ≤ 4	15	8.98	8.42	8.06
5	T-stat (max)	IF Central Perk ≤ 0.0537 AND exchanges_CHANDLER_JOEY ≤ 18.6667 AND exchanges_RACHEL_CHANDLER ≤ 4	15	8.98	8.42	8.06
1	T-stat (min)	IF exchanges_CHANDLER_JOEY ≤ 31.6667 AND exchanges_MONICA_CHANDLER ≤ 4 AND exchanges_ROSS_CHANDLER > 7	18	8.10	8.49	10.73
2	T-stat (min)	IF Main apartments ≤ 0.4667 AND exchanges_MONICA_CHANDLER ≤ 4 AND exchanges_ROSS_CHANDLER > 7	15	8.09	8.48	10.40
3	T-stat (min)	IF exchanges_CHANDLER_JOEY ≤ 18.6667 AND exchanges_MONICA_CHANDLER ≤ 4 AND exchanges_ROSS_CHANDLER > 7	16	8.12	8.48	10.36
4	T-stat (min)	IF Main apartments ≤ 0.5333 AND exchanges_MONICA_CHANDLER ≤ 4 AND exchanges_ROSS_CHANDLER > 7	19	8.13	8.49	9.04
5	T-stat (min)	IF Main apartments ≤ 0.4667 AND exchanges_MONICA_CHANDLER ≤ 4 AND exchanges_ROSS_CHANDLER > 4	18	8.12	8.49	8.42
1	BKW (max)	IF Central Perk ≤ 0.1176 AND character_lines_OTHER ≤ 0.1557	13	9.08	8.42	0.43
2	BKW (max)	IF Central Perk ≤ 0.1176 AND character_lines_OTHER ≤ 0.1557 AND exchanges_RACHEL_CHANDLER ≤ 4	26	8.91	8.40	0.42
3	BKW (max)	IF Central Perk ≤ 0.1176 AND exchanges_MONICA_PHOEBE ≤ 13.6667	10	9.13	8.43	0.41
4	BKW (max)	IF Central Perk ≤ 0.1176 AND exchanges_ROSS_PHOEBE > 2	14	9.03	8.42	0.41
5	BKW (max)	IF Central Perk ≤ 0.1176 AND character_lines_RACHEL > 0.1062	14	9.02	8.42	0.40
1	BKW (min)	IF character_lines_PHOEBE ≤ 0.103 AND humor > 0.634 AND vocab_richness > 0.4079	11	7.93	8.48	0.34
2	BKW (min)	IF Central Perk > 0.1667 AND character_lines_JOEY ≤ 0.1292 AND vocab_richness > 0.3865	28	8.10	8.51	0.34
3	BKW (min)	AND character_lines_PHOEBE ≤ 0.103 AND vocab_richness > 0.4079	10	7.91	8.48	0.34
4	BKW (min)	IF Central Perk > 0.0537 AND character_lines_PHOEBE ≤ 0.103 AND vocab_richness > 0.4079	12	7.96	8.49	0.33
5	BKW (min)	IF Central Perk > 0.1667 AND character_lines_JOEY ≤ 0.1492 AND vocab_richness > 0.3865	36	8.15	8.51	0.33
1	KLCSW (max)	IF Central Perk ≤ 0.1667 AND emotion_disgust ≤ 0.0534 AND exchanges_MONICA_CHANDLER > 4	68	8.69	8.36	0.33
2	KLCSW (max)	IF Central Perk ≤ 0.1667 AND emotion_disgust ≤ 0.0534 AND exchanges_MONICA_RACHEL > 2	63	8.70	8.37	0.32
3	KLCSW (max)	AND exchanges_MONICA_CHANDLER > 4 AND exchanges_RACHEL_CHANDLER ≤ 4	65	8.69	8.37	0.32
4	KLCSW (max)	IF Central Perk ≤ 0.1176 AND emotion_anger > 0.0421 AND exchanges_MONICA_CHANDLER > 4	62	8.70	8.37	0.32
5	KLCSW (max)	IF Central Perk ≤ 0.1176 AND character_lines_ROSS > 0.1034 AND emotion_disgust ≤ 0.0607	56	8.73	8.37	0.32
1	KLCSW (min)	IF Central Perk > 0.1667 AND emotion_anger ≤ 0.0793 AND exchanges_CHANDLER_JOEY ≤ 31.6667	66	8.24	8.54	0.30
2	KLCSW (min)	IF Central Perk > 0.1176 AND emotion_anger ≤ 0.0793 AND exchanges_ROSS_JOEY > 2	80	8.25	8.56	0.30
3	KLCSW (min)	IF Central Perk > 0.1667 AND emotion_anger ≤ 0.0793 AND exchanges_ROSS_JOEY > 2	56	8.21	8.54	0.30
4	KLCSW (min)	IF Central Perk > 0.1667 AND Hallways ≤ 0.0769 AND emotion_anger ≤ 0.0793	66	8.25	8.54	0.29
5	KLCSW (min)	IF Central Perk > 0.1667 AND Hallways ≤ 0 AND exchanges_CHANDLER_JOEY ≤ 31.6667	68	8.25	8.54	0.29

Note: Q stands for the quality score.

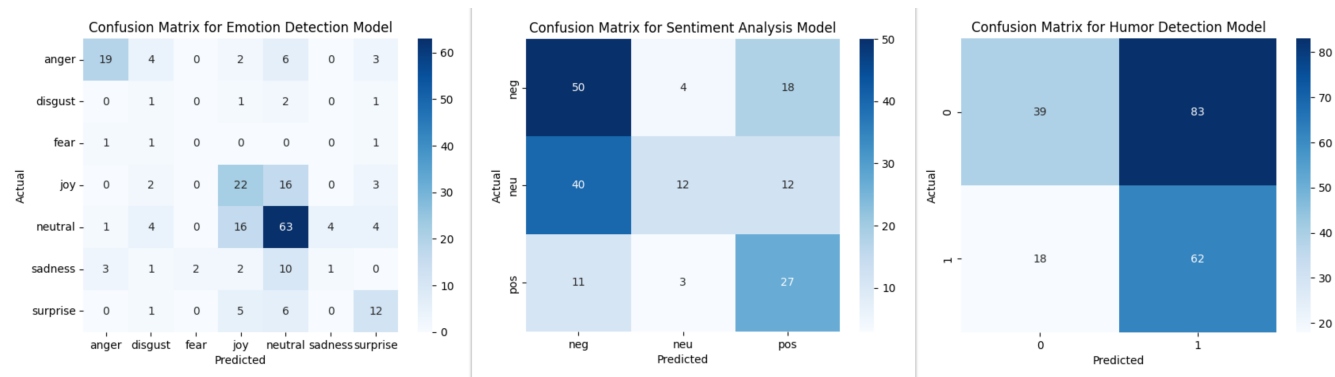


Figure 2: Confusion Matrices for the Emotion Detection, Sentiment Analysis, Humor Detection Models respectively.